Discrete Mathematics: Graph Theory revision

1 The vertices of a simple graph are numbered 1, 2, 3, 4, 5, 6. Two vertices are adjacent if the sum of their values is a prime number.

(a) Draw the graph.

(b) Give an example of a Hamiltonian cycle in this graph.

(c) Give an example of an Eulerian trail in this graph.

(d) Explain why there is not an Eulerian circuit.

A vertex numbered 7 is now added to the graph. The rule for adjacency remains the same.

(e) Show that there exists a Hamiltonian path, but not a Hamiltonian cycle in the graph.

(f) Explain why there is no Eulerian trail in this graph.

2 **(a)** Explain why no graph with degree sequence 2, 2, 2, 2, 2 can satisfy the condition of Ore's Theorem.

(b) Explain why every graph with degree sequence 2, 2, 2, 2, 2 must contain a Hamiltonian cycle.

3 **(a)** Draw a simply connected graph with degree sequence 5, 4, 4, 3, 2, 1,1.

(b) Explain why there is no simply connected graph with degree sequence 6, 4, 4, 3, 2, 1, 1.

4 Explain why no tree can contain a Hamiltonian cycle.

5 A connected tree has six vertices. Draw examples in which the largest vertex degree is:

(a) as large as possible;

(b) as small as possible.

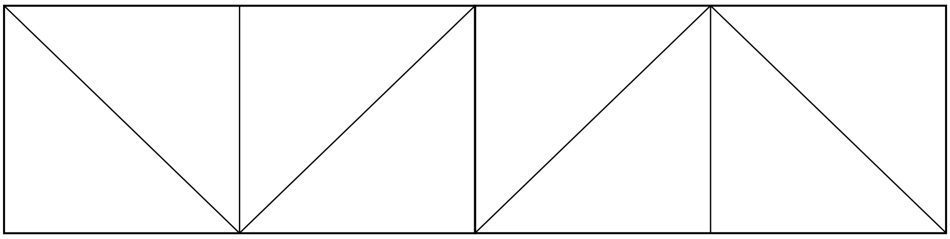
6 A simply connected graph has six vertices.

(a) What is the maximum number of edges in the graph?

(b) What is the maximum number of edges in the graph if it is also Eulerian?

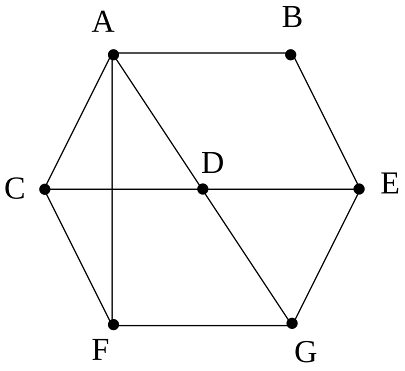
7 How many different spanning trees are there in a complete graph on four vertices?

8 If you want to trace the pattern below without taking your pencil off of the paper, then what is the minimum number of straight edges you will need to retrace?

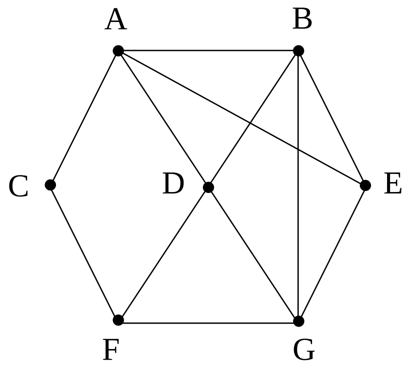


9 Which of the following graphs are isomorphic to each other? Explain how you decided.

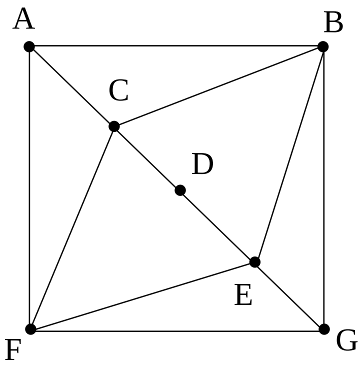
(a)



(b)



(c)



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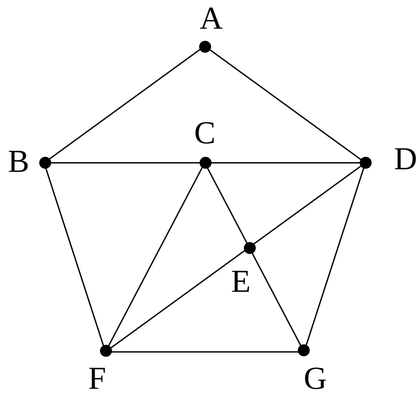
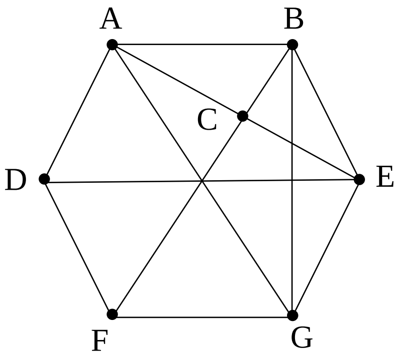
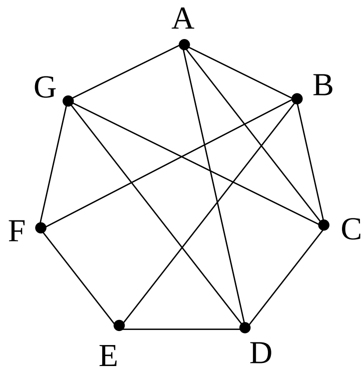


(e)



(f)



10 Which of the graphs in Question 9 are:

(a) Eulerian;

(b) Semi-Eulerian;

(c) Contain a Hamiltonian cycle;

(d) Satisfy the condition of Ore’s Theorem?

11 Consider graph (a) from Question 9. What is the minimum number of edges that need to be removed from this graph to make it bipartite?

12 **(a)** Arthur the Ant is initially at the vertex of a cube. He likes to travel to adjacent vertices by travelling along an edge of the cube.

(i) Show that it is possible for Arthur to visit every vertex of the cube once and only once and return to his original vertex at the end of his journey.

(ii) Explain why it isn’t possible for Arthur to travel along every edge of the cube exactly once.

(b) Silas the spider is initially in the middle of a face of a cube. He likes to travel to adjacent faces of the cube by crossing an edge of the cube.

(i) Show that it is possible for Silas to visit every face of the cube once and only once and return to his original face at the end of his journey.

(ii) Show that it is also possible for Silas to cross every edge of the cube exactly once.

13 Nine lines are drawn on a plane. Show that it is not possible for each line to intersect exactly three other lines.